

Diagonalization II

$T: V \longrightarrow V$ linear transformation

and suppose that T is diagonalizable

Therefore, we can find a basis

$$B = (v_1, v_2, \dots, v_n)$$

and $\lambda_1, \lambda_2, \dots, \lambda_n \in F$ s.t.

$$T(v_i) = \lambda_i v_i, \quad 1 \leq i \leq n$$

Observation (I)

In this situation, for any $m \geq 1$,

T^m is also diagonalizable.

In fact, $T^m(v_i) = \lambda_i^m \cdot v_i, \quad 1 \leq i \leq n$

In other words

$$[T^m]_{\mathcal{B}} = \begin{bmatrix} \lambda_1^m & & 0 \\ & \ddots & \\ 0 & & \lambda_n^m \end{bmatrix}$$

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$$[T]_{\mathcal{B}}^m$$

Example

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$$

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$$T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$e_1 \mapsto \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$e_2 \mapsto \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} \mapsto a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

" " "

$$ae_1 + be_2 \quad \begin{bmatrix} a+b \\ 2a+3b \end{bmatrix}$$

$$\begin{aligned} T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) &= A \begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} a+b \\ 2a+3b \end{bmatrix} \end{aligned}$$

Q: Is there a basis (v_1, v_2) for $\mathbb{R}^2$
 s.t. $T(v_1) = \lambda_1 v_1 \quad \exists \lambda_1, \lambda_2 \in \mathbb{R}$?
 $T(v_2) = \lambda_2 v_2$

$$\uparrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{bmatrix} 2a \\ 2b \end{bmatrix}$$

(2 unknown!)

$$\begin{bmatrix} a+b \\ 2a+3b \end{bmatrix}$$

Solve:

$$\begin{cases} a+b = 2a \\ 2a+3b = 2b \end{cases}$$

$$\Leftrightarrow (1-2)a + b = 0$$

$$2a + (3-2)b = 0$$

$$\begin{aligned} \text{det} &= (1-2)(3-2) \\ &= -2 \end{aligned}$$

$$\Leftrightarrow \left[\begin{array}{cc|c} 1-2 & 1 & 0 \\ 2 & 3-2 & 0 \end{array} \right]$$

(2 unknown!)

For this system to admit a

non-zero solution, the REF must

have a free variable

$$\begin{bmatrix} 1-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 2 & 3-\lambda \\ 1-\lambda & 1 \end{bmatrix}$$

$\downarrow \frac{1}{2} \times R_1$

$$\begin{bmatrix} 1, (3-\lambda)/2 \\ 0, \boxed{1 - \frac{(1-\lambda)(3-\lambda)}{2}} \end{bmatrix}$$

$R_2 - (1-\lambda) \cdot R_1$

$$\begin{bmatrix} \boxed{1} & (3-\lambda)/2 \\ 1-\lambda & 1 \end{bmatrix}$$

↑
must be 0
in order to
have a free
variable.

$$1 - \frac{(1-\lambda)(3-\lambda)}{2} = 0$$

$$\Leftrightarrow 2 = (1-\lambda)(3-\lambda)$$

$$\Leftrightarrow 2 = \lambda^2 - 4\lambda + 3$$

$$\Leftrightarrow \boxed{\lambda^2 - 4\lambda + 1 = 0}$$

$$\lambda = \frac{4 \pm \sqrt{16-4}}{2}$$

$$= 2 \pm \sqrt{3}$$

$$\lambda_1 = 2 + \sqrt{3}, \quad \lambda_2 = 2 - \sqrt{3}$$

$$a + b = (2 + \sqrt{3})a$$

$$2a + 3b = (2 + \sqrt{3})b$$

$$\Rightarrow \begin{bmatrix} 1 & \frac{3 - (2 + \sqrt{3})}{2} & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\Rightarrow a + \left(\frac{1 - \sqrt{3}}{2}\right)b = 0$$

$$\Rightarrow 2a + (1 - \sqrt{3})b = 0$$

Soln:

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sqrt{3} - 1 \\ 2 \end{bmatrix} = v_1$$

Similarly,

$$a + b = (2 - \sqrt{3})a$$

$$2a + 3b = (2 - \sqrt{3})b$$

$$\Leftrightarrow \begin{bmatrix} 1 & \frac{3 - (2 - \sqrt{3})}{2} & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\Leftrightarrow a + \left(\frac{1 + \sqrt{3}}{2}\right)b = 0$$

Soln:

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -(\sqrt{3} + 1) \\ 2 \end{bmatrix} = v_2$$

Then:

$$T(v_1) = (2 + \sqrt{3})v_1$$

$$T(v_2) = (2 - \sqrt{3})v_2$$

Since v_1 & v_2 are linearly independent,

$\mathcal{O} = (v_1, v_2)$ is a basis

showing that T is diagonalizable

Defⁿ The determinant of

a 2×2 -matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = A$

is the scalar

$$\det(A) = ad - bc$$

Proposition:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(F)$$

TF AE:

(1) A is invertible

(2) The columns of A are linearly independent

(3) The columns of A span F^2

(4) The columns of A are a basis for F^2

(5) $\det(A) \neq 0$

(6) $\text{RREF}(A) = I_2$

Pf: Only new thing here
is equivalence with 5

$$\boxed{(5) \Rightarrow (2)}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{matrix} (5) \\ \updownarrow \\ ad - bc \neq 0 \end{matrix}$$

Want to show: Columns are linearly
independent.

$$\text{If } \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} \lambda a \\ \lambda c \end{bmatrix}$$

$$\begin{aligned} \text{then } ad - bc &= a(\lambda c) - (\lambda a)c \\ &= 0 \end{aligned}$$

Can also argue

$$\boxed{(2) \Rightarrow (5)}$$

